RELEVANT PROJECT DOCUMENTATION

Online Food Ordering and Management System

1. System Design Document

The System Design Document explains how the Online Food Ordering and Management System will be designed and how the different parts of the system will work together.

1.1 System Architecture

The system will use a web-based structure with separate parts for the user interface, application logic, and database.

The main parts are:

- Customer Interface
- Restaurant Staff Interface
- Administrator Interface
- Application/Server Side
- Database

1.2 Customer Module

The Customer Module allows customers to:

- Create an account
- Log in
- View food items
- Search food items
- View food details
- Add food items to the cart
- Change item quantity
- Remove items from the cart
- Enter delivery details
- Select a payment method
- Place an order
- View order confirmation
- Check order status
- View previous orders
- Manage account details

1.3 Restaurant Staff Module

The Restaurant Staff Module allows restaurant staff to:

- Log in
- Add food items
- Edit food items
- Delete food items
- Update food prices
- Update food availability
- View customer orders
- View order details
- Update order status

1.4 Administrator Module

The Administrator Module allows the administrator to:

- Log in
- View users
- Manage user accounts
- Manage user roles and access
- View orders
- Manage system-level information

1.5 Main System Flow

Customer:

Register/Login → View Menu → Select Food → Add to Cart → Enter Delivery Details → Select Payment Method → Place Order → View Order Status

Restaurant Staff:

Login → View Orders → Check Order → Prepare Order → Update Order Status

Administrator:

Login → Manage Users/Access → View System Information → Manage System

2. Database Design Document

The Database Design Document explains the data required by the system and how the data is stored.

2.1 Main Tables

The system will contain the following main tables:

Users

Stores user account information.

Main fields:

- User ID
- Name
- Email
- Password
- Phone
- Role
- Account Status

Customers

Stores customer-related information.

Main fields:

- Customer ID
- User ID
- Address
- City

Food Items

Stores food item information.

Main fields:

- Food Item ID
- Food Name
- Description
- Price
- Category ID
- Availability
- Image

Categories

Stores food categories.

Main fields:

- Category ID
- Category Name
- Description

Orders

Stores customer order information.

Main fields:

- Order ID
- Customer ID
- Order Date
- Delivery Address
- Payment Method
- Order Status
- Total Amount

Order Items

Stores the food items included in each order.

Main fields:

- Order Item ID
- Order ID
- Food Item ID
- Quantity
- Unit Price
- Sub Total

Payments

Stores payment information.

Main fields:

- Payment ID
- Order ID
- Payment Method

- Payment Status
- Payment Date
- Amount

2.2 Main Relationships

- One user can have one customer account.
- One customer can place many orders.
- One order can contain many order items.
- One food item can appear in many order items.
- One category can contain many food items.
- An order can have payment information.

3. User Interface Design Document

The User Interface Design Document describes the main screens of the system.

3.1 Customer Screens

The customer side will include:

1. Home Page
2. Registration Page
3. Login Page
4. Food Menu Page
5. Food Details Page
6. Search Results Page
7. Cart Page
8. Checkout Page
9. Payment Selection Page
10. Order Confirmation Page
11. Order History Page
12. Order Status Page
13. Customer Profile Page

3.2 Restaurant Staff Screens

The restaurant staff side will include:

1. Staff Login
2. Dashboard
3. Food Item Management
4. Add Food Item
5. Edit Food Item

6. Food Availability Management
7. Order Management
8. Order Details
9. Order Status Update

3.3 Administrator Screens

The administrator side will include:

1. Admin Login
2. Admin Dashboard
3. User Management
4. Role/Access Management
5. Order View
6. System Information Management

3.4 Basic UI Rules

The system should:

- Use clear labels.
- Use readable text.
- Keep buttons easy to understand.
- Show error messages when incorrect data is entered.
- Show confirmation messages after important actions.
- Use the same layout style across the system.
- Make important information easy to find.

4. Use Case Documentation

Use cases describe how each user interacts with the system.

UC-01: Customer Registration

Actor: Customer

Purpose: Create a customer account.

Main Steps:

1. Customer opens the registration page.
2. Customer enters required details.
3. System checks the entered data.
4. System creates the account.

5. System shows a registration success message.

Result: Customer account is created.

UC-02: Customer Login

Actor: Customer

Purpose: Access the customer account.

Main Steps:

1. Customer enters email and password.
2. System checks the login details.
3. System allows access if the details are correct.
4. System shows the customer dashboard/home page.

Result: Customer is logged in.

UC-03: Place Order

Actor: Customer

Purpose: Place a food order.

Main Steps:

1. Customer views the menu.
2. Customer selects food items.
3. Customer adds items to the cart.
4. Customer checks the cart.
5. Customer enters delivery details.
6. Customer selects a payment method.
7. Customer confirms the order.
8. System creates the order.
9. System displays order confirmation.

Result: Order is created successfully.

UC-04: Manage Food Items

Actor: Restaurant Staff

Purpose: Manage food items.

Main Steps:

1. Staff logs in.
2. Staff opens food management.
3. Staff adds, edits, or deletes food items.
4. System validates the information.
5. System saves the changes.

Result: Food information is updated.

UC-05: Manage Orders

Actor: Restaurant Staff

Purpose: Manage customer orders.

Main Steps:

1. Staff logs in.
2. Staff opens the order management page.
3. Staff selects an order.
4. Staff views order details.
5. Staff updates the order status.
6. System saves the new status.

Result: Order status is updated.

UC-06: Manage Users

Actor: Administrator

Purpose: Manage system users.

Main Steps:

1. Administrator logs in.
2. Administrator opens user management.
3. Administrator views users.
4. Administrator manages user account/access information.
5. System saves the changes.

Result: User information/access is updated.

5. Test Plan

The Test Plan explains how the system will be tested.

5.1 Testing Objectives

Testing will be used to check whether:

- Functions work correctly.
- Data is saved correctly.
- Invalid data is rejected.
- User roles work correctly.
- Modules work together correctly.
- Orders are processed correctly.
- Security controls work correctly.

5.2 Main Testing Types

Unit Testing

Tests individual functions or small parts of the system.

Example:

Testing whether the **Add to Cart** function works correctly.

Integration Testing

Tests whether different modules work correctly together.

Example:

Customer → Order → Restaurant Staff

The order created by the customer should appear correctly in the restaurant staff order management module.

System Testing

Tests the complete system.

Example:

Login → Select Food → Cart → Checkout → Order → Order Status

User Acceptance Testing

Checks whether the completed system meets the required user functions.

6. Test Case Documentation

Test ID	Function	Test Action	Expected Result
TC-01	Registration	Enter valid customer details	Account is created
TC-02	Registration	Leave required field empty	Error message is shown
TC-03	Login	Enter correct email and password	User is logged in
TC-04	Login	Enter incorrect password	Login error is shown
TC-05	Food Search	Search for an existing food item	Matching food is displayed
TC-06	Cart	Add available food	Food is added to cart
TC-07	Cart	Enter quantity as 0	Validation error is shown
TC-08	Checkout	Enter valid delivery details	Checkout continues
TC-09	Order	Place valid order	Order is created
TC-10	Order	Try to order unavailable food	System prevents the order
TC-11	Staff	Add food item with valid data	Food item is saved
TC-12	Staff	Enter invalid food price	Validation error is shown
TC-13	Staff	Update order status	New status is saved
TC-14	Admin	View users	User list is displayed
TC-15	Security	Customer tries to access admin page	Access is denied

7. Data Validation Document

The system will validate user input before saving data.

7.1 Customer Information

The system should check:

- Name is not empty.
- Email is in a valid format.
- Phone number follows the required format.
- Password is required.
- Required delivery information is entered.

7.2 Food Information

The system should check:

- Food name is required.
- Category is required.
- Price must be a positive number.
- Availability must have a valid value.
- Required food information cannot be empty.

7.3 Order Information

The system should check:

- Customer must be logged in.
- Cart must contain at least one food item.
- Quantity must be greater than 0.
- Food must be available.
- Delivery details must be entered.
- Payment method must be selected.

8. Security Documentation

The system will include basic security controls.

8.1 Login Security

Users must log in before accessing protected functions.

8.2 Password Security

Passwords should not be stored as plain text. Passwords should be stored using a suitable password hashing method.

8.3 Role-Based Access

Different users should have access to different modules.

For example:

- Customer → Customer functions
- Restaurant Staff → Restaurant management functions
- Administrator → System management functions

8.4 Input Security

The system should validate user input to reduce invalid or unsafe data.

8.5 Unauthorized Access

A user should not be allowed to directly access functions that belong to another role.

Example:

A customer should not be able to open the administrator user-management page.

9. Integration Documentation

Integration documentation explains how the system modules communicate with each other.

9.1 Customer and Order Module

When a customer places an order:

Customer Module → Order Module

Customer information, selected food items, quantities, delivery details, and payment method are sent to the order process.

9.2 Order and Restaurant Module

After an order is created:

Order Module → Restaurant Staff Module

Restaurant staff can view the new order and update its status.

9.3 Order Status

The order status can move through stages such as:

Pending → Confirmed → Preparing → Ready → Delivered

If an order is cancelled, its status can become:

Cancelled

9.4 Database Integration

All modules that need shared information should use the same database structure.

For example:

The food selected by a customer must match the food information stored in the FoodItems table.

10. Error Handling Documentation

The system should display clear messages when an error occurs.

Example 1: Empty Field

Error:
"Please enter your email."

Example 2: Invalid Email

Error:
"Please enter a valid email address."

Example 3: Invalid Quantity

Error:
"Quantity must be greater than 0."

Example 4: Invalid Price

Error:
"Please enter a valid positive price."

Example 5: Wrong Login

Error:
"Invalid email or password."

Example 6: Unavailable Food

Error:
"This food item is currently unavailable."

Example 7: Unauthorized Access

Error:
"You do not have permission to access this page."

11. Change Management Document

Changes to the project requirements should be recorded before they are added to the system.

Change Process

1. Identify the requested change.
2. Record the change.
3. Check how the change affects the project.
4. Discuss the change with the supervisor if required.
5. Approve or reject the change.
6. Update the requirements document.
7. Update the design if needed.
8. Implement the change.
9. Test the change.
10. Update the project documents.

Example

If a new payment method is requested:

Request → Review → Approval → Requirements Update → Development → Testing → Documentation Update

12. Risk Documentation

Risk	Possible Effect	Response
Requirement changes	Extra development work	Record and review changes
Coding errors	System functions may fail	Test and fix errors
Database errors	Data may be lost or incorrect	Use backups and testing
Integration errors	Modules may not work together	Perform integration testing
Time shortage	Some functions may not be completed	Follow the project schedule
Security problems	Unauthorized access	Use authentication and access control
Internet problems	Development/testing delays	Continue offline work where possible
Payment problems	Online payment may fail	Test payment process carefully
Data loss	Project work may be lost	Keep backups and use Git
Testing delays	Errors may remain	Test each module during development

13. Version Control Documentation

Git and GitHub can be used to manage the project source code and document changes.

Version Control Rules

- Keep the project source code in the GitHub repository.
- Commit changes regularly.
- Use clear commit messages.
- Keep important project files in the repository.
- Avoid uploading passwords or private information.
- Keep backup copies of important project files.

Example Commit Messages

- Added customer registration
- Added food menu
- Added cart function
- Added order management
- Fixed login validation
- Updated database tables
- Added order status
- Fixed checkout error

14. User Manual

The User Manual explains how users can use the system.

14.1 Customer

Step 1: Register

Create a customer account by entering the required information.

Step 2: Login

Enter the registered email and password.

Step 3: View Food

Open the food menu and view available food items.

Step 4: Add Food to Cart

Select a food item and add it to the cart.

Step 5: Check Cart

Review the selected food items and quantities.

Step 6: Checkout

Enter the required delivery information.

Step 7: Select Payment

Select:

- Cash on Delivery
- Online Payment

Step 8: Place Order

Check the order details and confirm the order.

Step 9: Check Order Status

Open the order status page to see the current status.

14.2 Restaurant Staff

1. Log in.
2. Open the staff dashboard.
3. Manage food items.
4. Update food price or availability.
5. View customer orders.
6. Open order details.
7. Update the order status.

14.3 Administrator

1. Log in.
2. Open the administrator dashboard.
3. View users.

4. Manage user accounts and access.
5. View system orders.
6. Manage required system information.

15. Deployment Documentation

The Deployment Document explains the basic steps required to make the system available for use.

Deployment Steps

1. Prepare the final source code.
2. Prepare the production database.
3. Configure database connection settings.
4. Configure the application.
5. Test the deployed system.
6. Check login and user access.
7. Test customer ordering.
8. Test restaurant order management.
9. Test administrator functions.
10. Fix deployment errors.
11. Make the system available to users.

16. Maintenance Documentation

After the system is completed, maintenance may be required.

Maintenance activities include:

- Fixing errors.
- Updating food information.
- Updating system information.
- Updating security settings.
- Backing up the database.
- Updating system functions when required.
- Checking system performance.
- Updating documentation when changes are made.

17. Final Project Documentation Checklist

Before final submission, the following documents should be checked:

- [ ] Project Proposal
- [ ] Project Planning Document
- [ ] Requirements Documentation
- [ ] System Design Document
- [ ] Database Design Document
- [ ] User Interface Design
- [ ] Use Case Documentation
- [ ] Test Plan
- [ ] Test Cases
- [ ] Test Results
- [ ] Data Validation Documentation
- [ ] Security Documentation
- [ ] Integration Documentation
- [ ] Error Handling Documentation
- [ ] Change Management Document
- [ ] Risk Documentation
- [ ] Version Control Documentation
- [ ] User Manual
- [ ] Deployment Documentation
- [ ] Maintenance Documentation

- [ ] Final Project Report
- [ ] Source Code
- [ ] Database
- [ ] GitHub Repository

18. Document Control

All project documents should be kept updated during development.

Document	Purpose
Project Planning	Plans project work
Requirements	Defines system requirements
System Design	Defines system structure
Database Design	Defines database structure
UI Design	Defines system screens
Use Cases	Defines user interactions
Test Plan	Defines testing work
Test Cases	Defines test conditions
Security Document	Defines security controls
Integration Document	Defines module connections
User Manual	Explains system usage

Deployment Document	Explains system deployment
Maintenance Document	Explains future maintenance

19. Conclusion

These documents support the development of the Online Food Ordering and Management System from planning to final completion. They describe the system requirements, design, database, user interface, testing, security, integration, errors, risks, version control, deployment, and maintenance.

The documents should be updated when approved changes are made to the project. This helps keep the project information consistent with the actual system.